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GUIDANCE DOCUMENT 5F: SECTOR-SPECIFIC GUIDANCE DOCUMENT ON ELECTRICITY AS A GOOD

Disclaimer: This guidance document is not legally binding and it is of an explanatory nature. Legal provisions of Union law take precedence over the contents of this document and should always be consulted. The authentic texts of the EU legal instruments are those published in the Official Journal of the European Union. This guidance was prepared by Commission services and does not represent an official position of the Commission. Only the Court of Justice of the European Union is competent to authoritatively interpret Union law.

VERSION HISTORY

Date	Version notes
14.08.2026	First Publication of this document for the definitive period.

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1 PREFACE

1.1 Structure and content of the guidance documents

This series of documents has been written to support stakeholders by explaining the requirements of the CBAM Regulation in a non-legislative language. This series of guidance documents should benefit a variety of stakeholders. A specific focus has been put on clarifying the **requirements for operators of installations producing CBAM goods outside of the EU for the definitive period**.

- **Guidance 1** provides an overview of the main changes to the CBAM methodology between the definitive period and the concluded transitional period. It also introduces the CBAM and an overview of the compliance cycle, roles and responsibilities, as well as milestones and deadlines for operators of installations outside the EU during the definitive period and explains the general exemptions from the CBAM.

- **Guidance 2** provides a quick guidance for the intended reader of this document, the operator of an installation producing CBAM goods. It provides a roadmap of the most important concepts of CBAM emissions monitoring and where to find more information in this series of documents.

- **Guidance 3** sets out the monitoring and reporting obligations and recommendations which are potentially applicable to any affected producer of CBAM goods.

- **Guidance 4** provides an overview of how the adjustment for free allocation is to be determined and what information is needed.

- **Guidance documents 5a, 5b, 5c, 5d, 5e and 5f** present an overview per sector of the production processes and value chains for goods that are included in the scope of the CBAM, as well as sector-specific monitoring and reporting considerations for each CBAM good where this is relevant. These sections are supplemented by examples for each sector.

The series of guidance documents is accompanied by an electronic template for information that should be used by installation operators to communicate information to the authorised CBAM declarants and CBAM Registry.

Presentation of numbers in EU documents

To align with EU legal documents, this guidance document uses the following convention when presenting numbers.

The decimal separator used to separate the integral part of a number from its fractional part is a comma, e.g.: 0,890

Thousands, and powers of 10^{3n} thereafter, are separated by a space, e.g.:

- fifteen thousand is written as 15 000
- fifteen million is written as 15 000 000

1.2 Reference regulations and legal acts

Several regulations and legal acts are referred to throughout this series of guidance documents. For ease of reading and to avoid ambiguity, simplified references are used when citing these acts.

- **CBAM Regulation**: Regulation (EU) 2023/956 and Regulation (EU) 2025/2083 amending Regulation (EU) 2023/956. Notably, where article numbers are given in this document without further specification, they always refer to the CBAM Regulation.

  It can be accessed at the following link (consolidated version):
  http://data.europa.eu/eli/reg/2023/956/2025-10-20

- **Methodology Act:** Commission Implementing Regulation (EU) 2025/2547, in legal texts referred to as "Implementing act pursuant to Article 7(7) of the CBAM Regulation".

  It can be accessed at the following link:
  http://data.europa.eu/eli/reg_impl/2025/2547/oj

- **Default Values Act**: Commission Implementing Regulation (EU) 2025/2621, in legal texts referred to as "Implementing act pursuant to Article 7(7) of the CBAM Regulation", also referred to as "default values determined in accordance with Annex IV to the CBAM Regulation".

  It can be accessed at the following link: **http://data.europa.eu/eli/reg_impl/2025/2621/oj**

- **Verification Principles Act**: Commission Implementing Regulation (EU) 2025/2546, in legal texts referred to as "Implementing act pursuant to Article 8(3) of the CBAM Regulation".

  It can be accessed at the following link:
  http://data.europa.eu/eli/reg_impl/2025/2546/oj

- **Accreditation & Verification Requirements Act**: Commission Delegated Regulation (EU) 2025/2551, in legal texts referred to as "Delegated act pursuant to Article 18(3) of the CBAM Regulation".

  It can be accessed at the following link: **http://data.europa.eu/eli/reg_del/2025/2551/oj**

- **Free Allocation Adjustment Act**: Commission Implementing Regulation (EU) 2025/2620, in legal texts referred to as "Implementing act pursuant to Article 31(2) of the CBAM Regulation".

  It can be accessed at the following link:
  http://data.europa.eu/eli/reg_impl/2025/2620/oj

For acronyms and definitions used in this document, please see **Annex C** and **Annex B**.

A series of icons are used throughout to help guide the reader:

Icon	Description of use
	Points to information of particular importance for operators of installations producing CBAM goods.
Simplified!	Highlights simplified approaches of the general requirements of the CBAM.
	Used where recommended improvements are presented.
	Used where other documents, templates or electronic tools are available from other sources.
	Points to examples given for the topics discussed in the surrounding text.

2 ELECTRICITY "AS A GOOD" (I.E. IMPORTED INTO THE EU)

***Note**: The Commission's proposal[1] of 17 December 2025 to amend the CBAM Regulation includes proposed changes on the rules for electricity as CBAM good. However, at the time of writing this guidance document, the legislative discussion process is still ongoing. Therefore, this proposal is not reflected in this guidance document.*

The textbox below signposts sector-specific sections in the Methodology Act, relevant for the CBAM definitive period.

Methodology Act references:

- **Annex I, point 3** – Functional unit and system boundaries Sub-section **3.19** (Electricity)
- **Annex II, Section D** –Electricity, sub-sections **D.1** to **D.2**



Where electricity is imported into the EU as a good on its own, i.e. is not considered in the context of the indirect emissions of a (tangible) good, specific rules apply. Firstly, electricity is listed in Annex II to the CBAM Regulation and therefore only direct emissions are taken into account. Secondly, it is the exception from the rule that actual emissions are monitored instead of using a default value for the embedded emissions. For calculating those emissions, the following formula is used:

$$Em_{el} = E_{el} \cdot EF_{el} \qquad \text{(Equation 35)}$$

Where:

Em_{el} are the emissions related to electricity, expressed in t CO_2 ;

E_{el} is the electricity produced, expressed in MWh; and

EF_{el} is the emission factor for electricity applied, expressed in t CO_2/MWh.

For the emission factor of electricity, the rules given in section D.2 of Annex II to the Methodology Act have to be applied, which are explained below.

The following options for determining the emission factor of electricity apply:

(a) As the default case, the specific default value for a third country, group of third countries or region within a third country shall be used. That value is determined by the Commission based on the best data available to them. These **CO_2 emission factors** are based on data from the International Energy Agency (IEA) and are provided in Annex III to the Default Values Act.
(b) Where no specific default value is available pursuant to point (a), the CO_2 emission factor in the EU as set out in section D.2.2 of Annex II to the Methodology Act shall be used. It is also based on IEA data and provided in Annex III to the Default Values Act.
(c) Where a country, or group of third countries, submits sufficient evidence to demonstrate that the CO_2 emission factor in the third country, group of countries

[1] Proposal for a Regulation of the European Parliament and of the Council amending Regulation (EU) 2023/956 as regards the extension of its scope to downstream goods and anti-circumvention measures, COM/2025/989 final, https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A52025PC0989

or region within the third country from where the electricity is imported is lower than the values outlined in points (a) and (b), and where conditions provided in section D.2.3 of Annex II to the Methodology Act are met, the alternative default value will be used. However, as such an alternative value will be published by way of amendment of the Default Values Act, in practice there is no difference in approach for the authorised CBAM declarant.

(d) An authorised CBAM declarant may apply actual embedded emissions instead of default values, if it can be demonstrated that specific requirements provided in section 2.2 are met, and the calculation is based on data determined according to points D.4.1 or D.4.2 of Annex II to the Methodology Act.

2.1 CO_2 emission factor based on alternative reliable data

For the purpose of point (c) mentioned above, the third country, or group of third countries may propose the use of alternative default values for a given year if they provide, by 30 June of that year, the datasets from reliable alternative **official sources**, including national statistics, demonstrating that the CO_2 emission factor calculated as per equations 36 and 37 are lower than the default values laid down in the Default Values Act. If the Commission considers the alternative official sources to be reliable, it shall, where feasible, update the relevant default values by 30 June of the subsequent year. The Methodology Act contains the relevant calculation formulae (equations 36 and 37), building a five-year weighted average of all fossil electricity producers. The updated default values shall apply to electricity imports for the year in which the data from those sources were provided.

2.2 Elements of evidence for using actual embedded emissions for electricity imported into the Union.

Pursuant to point 5 of Annex IV to the CBAM Regulation, for allowing an authorised CBAM declarant to use actual emissions data of a specific electricity producing installation, all of the criteria (a) to (e) provided in section 5 of Annex IV to the CBAM Regulation have to be fulfilled, which are:

(a) The amount of electricity for which the use of actual embedded emissions is claimed is covered by a **power purchase agreement (PPA)** between the authorised CBAM declarant and a producer of electricity located in a third country;

(b) The installation producing electricity is **either directly connected to the Union transmission system or** it can be demonstrated that at the time of export there was **no physical network congestion** at any point in the network between the installation and the Union transmission system;

(c) The installation producing electricity **does not emit more than 550 grammes of CO_2** of fossil fuel origin **per kWh** of electricity;

(d) The amount of electricity for which the use of actual embedded emissions is claimed has been **firmly nominated to the allocated interconnection capacity** by all responsible transmission system operators in the country of origin, the country of destination and, if relevant, each country of transit, and the nominated capacity and the production of electricity by the installation refer to the same period of time, which shall not be longer than one hour.

(e) The fulfilment of the above criteria is certified by an accredited verifier, who shall receive at least monthly interim reports demonstrating how those criteria are fulfilled.

Furthermore, the said installation must determine the emission factor of electricity in line with Annex II to the Methodology

As clarified in section D.2.4 of Annex II to the Methodology Act, the following elements of evidence shall be provided to justify the use of actual emissions.

- For criterion (a):
    i. There must be contractual evidence demonstrating the existence of a power purchase agreement (PPA) concluded directly between the authorised CBAM declarant and a producer of electricity located in a third country for the physical delivery of electricity. The PPA shall be applicable at the time of electricity import for which actual emissions are claimed and shall cover at least the amount of electricity for which actual emissions are claimed. Where the PPA was concluded through an intermediary, the contractual evidence shall demonstrate that only one single contract was concluded between the three contracting parties.
- For criterion (b), one of the following:
    i. A single line diagram demonstrating the existence of a direct connection between the installation producing electricity and the Union transmission system;
    ii. Written documentation, either from the transmission system operator or from another entity with access to relevant information, attesting that at the time of export, determined on an hourly basis, there was no physical network congestion at any point in the network between the installation and the Union transmission system.
- For criterion (c):
    i. Data showing that the installation producing electricity does not emit more than 550 grammes of CO_2 of fossil fuel origin per kilowatt-hour of electricity.
- For criterion (d):
    i. Written documentation, either from the person who nominated the relevant capacity at the interconnector or from the relevant transmission system operator, demonstrating that a given quantity of electricity has been nominated in the country of origin, the country of destination and, if relevant, each country of transit, and demonstrating the period of time to which the nomination of capacity refers; and
    ii. Data from a smart metering system demonstrating that the production of a corresponding amount of electricity by the installation occurred within the same measurement period as the nomination of the capacity. This period shall not exceed one hour.
- For criterion (e):

i. Monthly interim reports containing the elements of evidence laid down in this point demonstrating how criteria (a) to (d) as laid down in point 5 of Annex IV to the CBAM Regulation are fulfilled.

The evidence for these criteria is to be provided to the verifier of the installation producing the electricity for which actual data are to be used. For making the evidence (or the verification thereof) accessible to the declarants importing electricity, the installation's emission report must be supplemented by "*Declarant-specific addendums*" as specified in section 1.1.1 of Annex IV to the Methodology Act, which have to contain:

1. The EORI number of the authorised CBAM declarant to whom the declarant-specific addendum refers;
2. An indication that the criteria for the use of actual values laid down in point 5, first subparagraph, points (a) and (d) of Annex IV to Regulation (EU) 2023/956, as well as, where relevant, laid down in point 5 first subparagraph, point (b) of Annex IV to that Regulation in relation to the lack of physical network congestion, are met, and a confirmation that the related elements of evidence laid down in point D.2.4 of Annex II were submitted to the verifier;
3. The quantity of electricity imported by that authorised CBAM declarant from the relevant installation for which the criteria laid down in point 5 of Annex IV to Regulation (EU) 2023/956 are met.

3 WORKED EXAMPLE FOR ELECTRICITY

The following example demonstrates how an installation that exports electricity to the EU can determine the electricity emissions factor. The example installation consists of two blocks: Block 1 is a relatively old coal-fired baseload power plant, which produces only electricity. It is assumed that after 2016 a modern gas-fired cogeneration (CHP) plant was added as Block 2. As Block 2 can react more easily to market demand for heat (district heating) and electricity, it runs for a smaller number of full-load hours, but still for a significant part of the year. Both blocks are monitored separately, but for CBAM purposes, the weighted average of the electricity emission factor is determined. The emissions then need to be verified and provided to the authorised CBAM declarant who imports the electricity into the EU.

For simplicity, standard values are used for the properties of fuels and emissions from flue gas cleaning are omitted in this example. Below, the separate calculations for each block, and thereafter for the whole installation are demonstrated.

1. Coal-fired baseload power plant

Capacity:		500	MW thermal input		
Electrical Efficiency:		35%			
Full-load operating hours:			7 800	h/year	
Coal:	Emission factor =		96,1	t CO_2 / TJ	
	NCV =		25,8	GJ/t	
Coal input:				544 186	t coal / year
Energy input from fuels:				14 040	TJ / year

Emissions:	1 349 244	t CO_2 / year
Electricity production:	1 365 000	MWh
EF electricity Block 1:	**0,988**	**t CO_2 / MWh**

2. Natural-gas-fired Cogeneration plant

Capacity:	850	MW thermal input	
Full load hours:	6 000	h/year	
Actual efficiency electricity:	49%		
Actual efficiency heat:	31%		
Natural gas EF =	56,1	t CO2 / TJ	
NCV =	48,0	GJ/t	
Natural gas input:	382 500	t / year	
Energy input from fuels:	18 360	TJ / year	
Emissions:	1 029 996	t CO2 / year	
Electricity production:	2 499 000	MWh	

For applying the CHP-related calculations (point A.2.2 of Annex III to the Methodology Act), the operator needs to determine the applicable reference efficiencies for separate heat and electricity production. Those are given in section C of Annex III to the Methodology Act, and in **Annex A** of this document. It is assumed that this CHP was built after 2015. Therefore, the following values for natural gas are determined:

Reference efficiency electricity	53%
Reference efficiency heat (hot water)	92%

The following calculation is required:

$$F_{CHP,El} = \frac{\eta_{el}/\eta_{ref,el}}{\eta_{heat}/\eta_{ref,heat}+\eta_{el}/\eta_{ref,el}} = \frac{0{,}49/0{,}53}{0{,}31/0{,}92+0{,}49/0{,}53} = 73{,}3\%$$

$$EF_{CHP,El} = Em_{CHP} \cdot F_{CHP,El} / E_{El,prod} = 1\,029\,996 \text{ t } CO_2 \times 0{,}733 / 2\,499\,000 \text{ MWh} =$$
$$= 0{,}3021 \text{ t } CO_2 / \text{MWh}$$

3. Calculation of the weighted average

The specific embedded (direct) emissions of electricity, which is the emission factor of electricity, of the whole installation is calculated as the weighted average of the emission factors of the two power plant blocks, as follows:

	Electricity produced (MWh)	**Emission factor (EF) [t CO_2 / MWh]**
Block 1	1 365 000	0,988
Block 2	2 499 000	0,302
Total installation:	3 864 000	**0,545**

Therefore, the emission factor to be used by an authorised declarant for electricity from this specific installation will be 0,545 t CO_2 / MWh. Note that this at the same time provided evidence for meeting criterion (c) of point 5 of Annex IV to the CBAM Regulation (see section 2.2).

Annex A Harmonised efficiency reference values for separate production of electricity and heat

Methodology Act: Annex III.

In the tables below the harmonised efficiency reference values for separate production of electricity and heat are based on net calorific value and standard atmospheric ISO conditions (15 °C ambient temperature, 1,013 bar, 60 % relative humidity).

Table 3-1: Reference efficiency factors for electricity production

Category		Type of fuel	Year of construction		
			Before 2012	2012-2015	From 2016
Solids	S1	Hard coal including anthracite, bituminous coal, sub-bituminous coal, coke, semi-coke, pet coke	44,2	44,2	44,2
	S2	Lignite, lignite briquettes, shale oil	41,8	41,8	41,8
	S3	Peat, peat briquettes	39,0	39,0	39,0
	S4	Dry biomass including wood and other solid biomass including wood pellets and briquettes, dried woodchips, clean and dry waste wood, nut shells and olive and other stones	33,0	33,0	37,0
	S5	Other solid biomass including all wood not included under S4 and black and brown liquor	25,0	25,0	30,0
	S6	Municipal and industrial waste (non-renewable) and renewable/bio-degradable waste	25,0	25,0	25,0
Liquids	L7	Heavy fuel oil, gas/diesel oil, other oil products	44,2	44,2	44,2
	L8	Bio-liquids including bio-methanol, bioethanol, bio-butanol, biodiesel, and other bio-liquids	44,2	44,2	44,2
	L9	Waste liquids including biodegradable and non-renewable waste (including tallow, fat and spent grain)	25,0	25,0	29,0
Gaseous	G10	Natural gas, LPG, LNG and biomethane	52,5	52,5	53,0
	G11	Refinery gases hydrogen and synthesis gas	44,2	44,2	44,2
	G12	Biogas produced from anaerobic digestion, landfill, and sewage treatment	42,0	42,0	42,0
	G13	Coke oven gas, blast furnace gas, mining gas, and other recovered gases (excluding refinery gas)	35,0	35,0	35,0
Other	O14	Waste heat (including high temperature process exhaust gases, product from exothermic chemical reactions)			30,0

Table 3-2: Reference efficiency factors for heat production

Category		Type of fuel	Year of construction					
			Before 2016			From 2016		
			Hot water	Steam ([2])	Direct use of exhaust gases ([3])	Hot water	Steam ([2])	Direct use of exhaust gases ([3])
Solids	S1	Hard coal including anthracite, bituminous coal, sub-bituminous coal, coke, semi-coke, pet coke	88	83	80	88	83	80
	S2	Lignite, lignite briquettes, shale oil	86	81	78	86	81	78
	S3	Peat, peat briquettes	86	81	78	86	81	78
	S4	Dry biomass including wood and other solid biomass including wood pellets and briquettes, dried woodchips, clean and dry waste wood, nut shells and olive and other stones	86	81	78	86	81	78
	S5	Other solid biomass including all wood not included under S4 and black and brown liquor	80	75	72	80	75	72
	S6	Municipal and industrial waste (non-renewable) and renewable/bio-degradable waste	80	75	72	80	75	72
Liquids	L7	Heavy fuel oil, gas/diesel oil, other oil products	89	84	81	85	80	77

([2]) If steam plants do not account for the condensate return in their calculation of CHP (combined heat and power) heat efficiencies, the steam efficiencies shown in the table above shall be increased by 5 percentage points.

([3]) Values for direct use of exhaust gases shall be used if the temperature is 250 °C or higher.

Category		Type of fuel	Year of construction					
			Before 2016			From 2016		
			Hot water	Steam ([2])	Direct use of exhaust gases ([3])	Hot water	Steam ([2])	Direct use of exhaust gases ([3])
	L8	Bio-liquids including bio-methanol, bioethanol, bio-butanol, biodiesel, and other bio-liquids	89	84	81	85	80	77
	L9	Waste liquids including biodegradable and non-renewable waste (including tallow, fat and spent grain)	80	75	72	75	70	67
Gaseous	G10	Natural gas, LPG, LNG and biomethane	90	85	82	92	87	84
	G11	Refinery gases hydrogen and synthesis gas	89	84	81	90	85	82
	G12	Biogas produced from anaerobic digestion, landfill, and sewage treatment	70	65	62	80	75	72
	G13	Coke oven gas, blast furnace gas, mining gas, and other recovered gases (excluding refinery gas)	80	75	72	80	75	72
Other	O14	Waste heat (including high temperature process exhaust gases, product from exothermic chemical reactions)	—	—	—	92	87	—

Annex B List of definitions

Term	Definition
'Accuracy'	means the closeness of the agreement between the result of a measurement and the true value of the particular quantity or a reference value determined empirically using internationally accepted and traceable calibration materials and standard methods, taking into account both random and systematic factors
'Activity data'	means the amount of fuels or materials consumed or produced by a process relevant for the calculation-based methodology, expressed in terajoules (TJ), mass in tonnes or (for gases) volume in normal cubic metres, as appropriate
'Actual emissions'	means the emissions calculated based on primary data from the production processes of goods and from the production of electricity consumed during those processes as determined in accordance with the methods set out in Annex IV [to the Methodology Act]
'Activity level'	means the quantity of goods to which the same functional unit applies that are produced within the system boundaries of a production process during a reporting period
'Aggregated goods categories'	means aggregated goods categories pursuant to Table 1 of point 2 of Annex I. A group of CBAM goods with their CN codes, used for the purpose of defining system boundaries of production processes.
'Agricultural, aquaculture, fisheries and forestry residues'	means residues that are directly generated by agriculture, aquaculture, fisheries and forestry and that do not include residues from related industries or processing
'Authorised CBAM declarant'	means a person authorised by the competent authority in accordance with Article 17 [of the CBAM Regulation]. The 'authorised CBAM declarant' [which replaces the term 'reporting declarant' from the transitional period] may be importers of CBAM goods, or their indirect customs representatives, who have applied for and been granted the status of authorised CBAM declarant.
'Batch'	means an amount of fuel or material representatively sampled and characterised, and transferred as one shipment or continuously over a specific period of time
'Biomass'	means biomass as defined in Article 2, point (24) of the Renewable Energy Directive (RED II)[4]; it includes bioliquids and biofuels as defined in Article 2, point (32) and (33), biomass fuels as defined in Article 2, point (27) and biogas as defined in Article 2 point (28) of the RED II;

[4] Directive (EU) 2018/2001

Term	Definition
'Biomass fraction'	means the ratio of carbon stemming from biomass to the total carbon content of a fuel or material, expressed as a fraction
'Calculation factors'	means net calorific value, emission factor, preliminary emission factor, oxidation factor, conversion factor, carbon content or biomass fraction
'Calibration'	means the set of operations, which establishes, under specified conditions, the relations between values indicated by a measuring instrument or measuring system, or values represented by a material measure or a reference material and the corresponding values of a quantity realised by a reference standard
'Carbon price'	means the monetary amount paid in a third country, under a carbon emissions reduction scheme, in the form of a tax, levy or fee or in the form of emission allowances under a greenhouse gas emissions trading system, calculated on greenhouse gases covered by such a measure, and released during the production of goods
'CBAM certificate'	means a certificate in electronic format corresponding to one tonne of CO_2e of embedded emissions in goods
'CO_2 emission factor'	means the weighted average of the CO_2 intensity of electricity produced from fossil fuels within a geographic area. The CO_2 emission factor is the result of the division of the CO_2 emission data of the electricity sector by the gross electricity generation based on fossil fuels in the relevant geographic area. It is expressed in tonnes of CO_2 per megawatt-hour
'Combined nomenclature' (CN)	means the classification of goods, designed to meet the needs of: i) the Common customs tariff, setting import duties for products imported into the European Union (EU), as well as the Integrated tariff of the European Communities (Taric), incorporating all EU and trade measures applied to goods imported into and exported out of the EU; ii) the international trade statistics of the EU. The CN provides the means of collecting, exchanging and publishing data on EU international trade statistics. It is also used for the collection and publication of international trade statistics in intra-EU trade.[5]
'Combustion emissions'	means greenhouse gas emissions occurring during the exothermic reaction of a fuel with oxygen
'Competent authority'	means the authority designated by each Member State in accordance with Article 11 [of the CBAM Regulation]

[5] For definition see: https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Glossary:Combined_nomenclature_(CN)

Term	Definition
'Continuous emission measurement' (CEM)	means a set of operations having the objective of determining the value of a quantity by means of periodic measurements, applying either measurements in the stack or extractive procedures with a measuring instrument located close to the stack, whilst excluding measurement methodologies based on the collection of individual samples from the stack
'Control system'	means the operator's risk assessment and entire set of control activities, including the continuous management thereof, that an operator has established, documented, implemented and maintained pursuant to point A.2. of Annex II
'Complex goods'	means goods other than simple goods
'Conservative'	means that a set of assumptions is defined in order to ensure that no under-estimation of reported emissions or over-estimation of production of heat, electricity or goods occurs
'Conversion factor'	means the ratio of carbon emitted as CO_2 to the total carbon contained in the source stream before the emitting process takes place, expressed as a fraction, considering CO emitted to the atmosphere as the molar equivalent amount of CO_2
'Co-product'	means any of two or more products resulting from the same production process
'Customs declarant'	means a declarant as defined in Article 5, point (15) of Regulation (EU) No 952/2013 lodging a customs declaration for release for free circulation of goods in its own name or the person in whose name such a declaration is lodged
'CCUS system'	means a group of economic operators with technically connected installations and transport equipment for CO_2 capture, transport, use in production of goods, or geological storage
'Data flow activities'	mean activities related to the acquisition, processing and handling of data that are needed to draft an emissions report from primary source data
'Data set'	means one type of data, either at installation level or production process level as relevant in the circumstances, as any of the following: (a) the amount of fuels or materials consumed or produced by a production process as relevant for the calculation-based methodology, expressed in terajoules, mass in tonnes, or for gases as volume in normal cubic metres, as appropriate, including for waste gases; (b) a calculation factor;

Term	Definition
	(c) net quantity of measurable heat, and the relevant parameters required for determining this quantity, in particular: i) mass flow of heat transfer medium; and ii) enthalpy of transmitted and returned heat transfer medium, as specified by composition, temperature, pressure and saturation; (d) quantities of non-measurable heat, specified by the relevant quantities of fuels used for producing the heat, and the net calorific value (NCV) of the fuel mix; (e) quantities of electricity; (f) quantities of CO_2 transferred between installations; (g) quantities of precursors received from outside the production process, and their relevant parameters, such as country of origin, used production route, specific direct and indirect emissions;
'Default value'	means a value, which is calculated or drawn from secondary data, which represents the embedded emissions in goods
'Direct emissions'	means emissions from the production processes of goods including emissions from the production of heating and cooling that is consumed during the production processes, irrespective of the location of the production of the heating or cooling
'Embedded emissions'	means direct emissions released during the production of goods and indirect emissions from the production of electricity that is consumed during the production processes, calculated in accordance with the methods set out in Annex IV and further specified in the implementing acts adopted pursuant to Article 7(7)
'Emissions'	mean the release of greenhouse gases into the atmosphere from the production of goods
'Emission factor'	means the average emission rate of a greenhouse gas relative to the activity data of a source stream assuming complete oxidation for combustion and complete conversion for all other chemical reactions
'Emission factor' for electricity	means the default value, expressed in CO_2e, representing the emission intensity of electricity consumed in production of goods.
'Emission source'	means a separately identifiable part of an installation or a process within an installation, from which relevant greenhouse gases are emitted
EU ETS	means the system for greenhouse gas emissions allowance trading within the Union in respect of activities listed in Annex I to Directive 2003/87/EC other than aviation activities

Term	Definition
'EU ETS allowance'	means an allowance as defined in Article 3, point (a), of Directive 2003/87/EC in respect of activities listed in Annex I to that Directive other than aviation activities
'Fossil carbon'	means inorganic and organic carbon that is not biomass
'Fossil fraction'	means the ratio of fossil carbon to the total carbon content of a fuel or material, expressed as a fraction
'Fugitive emissions'	means irregular or unintended emissions from sources that are not localised, or too diverse or too small to be monitored individually
'Functional unit'	means the reference unit used for the calculation of embedded emissions in goods
'Goods'	means goods listed in Annex I [to the CBAM the CBAM Regulation]
'Greenhouse gases'	mean greenhouse gases as specified in Annex I [to the CBAM the CBAM Regulation] in relation to each of the goods listed in that Annex
'Heat meter'	means a thermal energy meter or any other device to measure and record the amount of thermal energy produced based upon flow volumes and temperatures
'Importer'	means either the person lodging a customs declaration for release for free circulation of goods or a bill of discharge in accordance with Article 175(5) of Delegated Regulation (EU) 2015/2446 in its own name and on its own behalf or, where the customs declaration is lodged by an indirect customs representative in accordance with Article 18 of Regulation (EU) No 952/2013, the person on whose behalf such a declaration is lodged
'Importation'	means release for free circulation as provided for in Article 201 of Regulation (EU) No 952/2013
'Indirect customs Representative'	A customs representative appointed by an importer established in a Member State. If the importer appoints an indirect customs representative and they agree, they can act as an authorised CBAM declarant and submit the application for an authorisation on behalf of the importer. The indirect customs representative is liable for all operations made for the importers it represents
'Indirect emissions'	means emissions from the production of electricity which is consumed during the production processes of goods, irrespective of the location of the production of the consumed electricity
'Inherent CO_2'	means CO_2 which is part of a source stream.
'Installation'	means a stationary technical unit where a production process is carried out

Term	Definition
'Legal metrological control'	means the control exercised by a public authority or regulator of the measurement tasks intended for the field of application of a measuring instrument, for reasons of public interest, public health, public safety, public order, protection of the environment, the levying of taxes and duties, the protection of consumers and fair trading
'Measurable heat'	means a net heat flow transported through identifiable pipelines or ducts using a heat transfer medium, such as, in particular, steam, hot air, water, oil, liquid metals and salts, for which a heat meter is or could be installed
'Measurement point'	means the emission source for which continuous emission measurement systems (CEMS) are used for emission measurement, or the cross-section of a pipeline system for which the CO_2 flow is determined using continuous measurement systems
'Measurement system'	means a complete set of measuring instruments and other equipment used to determine variables for the monitoring and calculation of emissions
'Minimum requirements'	means monitoring methods using the minimum efforts allowed for determining data in order to result in emission data acceptable for the purpose of the CBAM Regulation.
'Mixed fuel'	means a fuel which contains both biomass and fossil carbon
'Mixed material'	means a material which contains both biomass and fossil carbon
'Multifunctional process'	means a process that delivers multiple outputs or whose outputs feed into several production processes
'Net calorific value' (NCV)	means the specific amount of energy released as heat when a fuel or material undergoes complete combustion with oxygen under standard conditions, less the heat of vaporisation of any water formed
'Non-CBAM good'	means any good produced in the installation that is not included in Annex I to the CBAM Regulation
'Non-measurable heat'	means all heat other than measurable heat
'Operator'	means any person who operates or controls an installation in a third country, including a parent company that controls an installation in a third country
'Oxidation factor'	means the ratio of carbon oxidised to CO_2 as a consequence of combustion to the total carbon contained in the fuel, expressed as a fraction, considering carbon monoxide (CO) emitted to the atmosphere as the molar equivalent amount of carbon dioxide (CO_2)
'Power purchase agreement'	means a contract under which a person agrees to purchase electricity directly from an electricity producer

Term	Definition
'Precursor'	means any input material into a production process included in the list of goods set out in Annex I to the CBAM Regulation;
'Preliminary emission factor'	means the assumed total emission factor of a fuel or material based on the carbon content of its biomass fraction and its fossil fraction before multiplying it by the fossil fraction to produce the emission factor
'Production process'	means the chemical and physical processes carried out to produce goods in an installation
'Production route' [6]	means a specific technology used in a production process to produce goods
'Process emissions'	means greenhouse gas emissions other than combustion emissions occurring as a result of intentional and unintentional reactions between substances or their transformation, for a primary purpose other than the generation of heat, including from the following processes: (a) the chemical, electrolytic or pyrometallurgical reduction of metal compounds in ores, concentrates and secondary materials; (b) the removal of impurities from metals and metal compounds; (c) the decomposition of carbonates, including those used for flue gas cleaning; (d) chemical syntheses of products and intermediate products where the carbon bearing material participates in the reaction; (e) the use of carbon containing additives or raw materials; (f) the chemical or electrolytic reduction of metalloid oxides or non-metal oxides such as silicon oxides and phosphates.
'Proxy data'	means annual values which are empirically substantiated or derived from accepted sources and which an operator uses to substitute a data set[7] for the purpose of ensuring complete reporting
'Rebate'	means any amount that reduces the amount due or paid by a person liable for the payment of a carbon price, before its payment or after, in a monetary form or in any other form.
'Recommended improvements'	means monitoring methods which are proven means to ensure that data are more accurate or less prone to mistakes than by mere application of minimum requirements
'Related emissions'	means: – the direct emissions caused by the source stream or emission source concerned; – emissions attributed to a quantity of measurable heat; – the indirect emissions related to the quantity of electricity concerned; – embedded emissions of a material produced or of a precursor consumed.

[6] Note that different production routes can fall within the same production process.

[7] Refers to the activity data or the calculation factors.

Term	Definition
'Reporting period'	means the period corresponding to the calendar year during which the good was produced and used by the authorised CBAM declarant as reference for the determination of embedded emissions
'Residue'	means a substance that is not the end product that a production process directly seeks to produce; it is not a primary aim of the production process and the process has not been deliberately modified to produce it
'Simple goods'	means goods produced in a production process requiring exclusively input materials (precursors) and fuels having zero embedded emissions
'Source stream'	means either of the following: (a) a specific fuel type, raw material or product giving rise to emissions of relevant greenhouse gases at one or more emission sources as a result of its consumption or production; (b) a specific fuel type, raw material or product containing carbon and included in the calculation of greenhouse gas emissions using a mass-balance method
'Specific embedded emissions'	means the embedded emissions of one tonne of goods, expressed as tonnes of CO_2e emissions per tonne of goods
'Standard conditions'	means temperature of 273,15 K and pressure conditions of 101 325 Pa defining normal cubic metres (Nm^3)
'System boundary'	means the group of chemical or physical processes included in the calculation of embedded emissions of goods under the same aggregated goods category. The system boundaries shall cover direct emissions, indirect emissions for goods not listed in Annex II to the CBAM Regulation, and the embedded emissions of any precursor.
'Third country'	means a country or territory outside the customs territory of the [European] Union
'Tonne of $CO_2(e)$'	means one metric tonne of carbon dioxide ('CO_2'), or an amount of any other greenhouse gas listed in Annex I [to the CBAM Regulation] with an equivalent global warming potential ['CO_2e']
'Transmission system operator'	means an operator as defined in Article 2, point (35) of Directive (EU) 2019/944 of the European Parliament and of the Council ([8]).

[8] Directive (EU) 2019/944 of the European Parliament and of the Council of 5 June 2019 on common rules for the internal market for electricity and amending Directive 2012/27/EU (OJ L 158, 14.6.2019, p. 125).

Term	Definition
'Uncertainty'	means a parameter, associated with the result of the determination of a quantity, that characterises the dispersion of the values that could reasonably be attributed to the particular quantity, including the effects of systematic as well as of random factors, expressed in percent, and describes a confidence interval around the mean value comprising 95% of inferred values taking into account any asymmetry of the distribution of values
'Waste'	means any substance or object which the holder discards or intends or is required to discard, excluding substances that have been intentionally modified or contaminated in order to meet this definition
'Waste gas'	means a gas containing incompletely oxidised carbon in a gaseous state under standard conditions which is a result of any of the processes listed under 'process emissions'

Annex C List of abbreviations

Abbreviation	Full term
AD	Activity Data
AEM	Anode Effect Minutes
AEO	Anode Effect Overvoltage
AL	Activity Level
AOD	Argon Oxygen Decarburisation.
BAT	Best Available Techniques
BF	Biomass Fraction
BFG	Blast Furnace Gas
BM	Benchmark
BOF	Basic Oxygen Furnace
BOFG	Basic Oxygen Furnace Gas
BREFs	Best Available Techniques Reference documents
CA	Competent Authority
CBAM	Carbon Border Adjustment Mechanism, the
CCR	Clinker to Cement Ratio
CCS	Carbon Capture and Storage
CCU(S)	Carbon Capture and Utilisation (and Storage)
CEMS	Continuous Emission Measurement Systems
CF	Conversion Factor
CFP	Carbon footprint of products
CHP	Combined Heat and Power
CKD	Cement Kiln Dust
CN	Combined nomenclature
COG	Coke Oven Gas
DRI	Direct Reduced Iron
EAF	Electric Arc Furnace
EEA	European Economic Area (EU-27, Iceland, Norway, Liechtenstein)
EF	Emission Factor
EFTA	European Free Trade Area
EORI	Economic Operator Registration and Identification
ETS	Emissions Trading System
EU ETS	EU Emissions Trading System
EUA	EU Allowances (used in the EU ETS)
EUR	Euro (currency)

Abbreviation	Full term
FAR	Free Allocation Rules (Regulation 2019/331)[9]
FIFO	First-In-First-Out principle
GHG	Greenhouse Gas
GWP	Global Warming Potential
HBI	Hot Briquetted Iron
HS	Harmonised System (for international trade)
IEA	International Energy Agency
ISO	International Organization for Standardization
LULUCF	Land-use, land change and forestry (criteria)
MP	Monitoring Plan
MRR	Monitoring and Reporting Regulation (Regulation 2018/2066)[10]
MRV	Monitoring, Reporting and Verification
MS	Member State(s)
MWh	Megawatt-hour
NCV	Net Calorific Value
NPI	Nickel pig iron
OF	Oxidation Factor
PCI	Pulverised Coal Injection
PEMS	Predictive Emission Monitoring System
PFC	Perfluoro-carbon
PoS	Proofs of Sustainability
RED II	Renewable Energy Directive, re-cast
SEE	Specific embedded emissions
SEFA	Specific embedded free allocation
TARIC	Integrated Tariff of the European Union database
TJ	Terajoules
TSO	Transmission System Operator
UCC	Union Custom Code
UN/LOCODE	United Nations Code for Trade and Transport Location

[9] Free Allocation Rules (Commission Delegated Regulation (EU) 2019/331 of 19 December 2018 determining transitional Union-wide rules for harmonised free allocation of emission allowances pursuant to Article 10a of Directive 2003/87/EC of the European Parliament and of the Council)

[10] Monitoring and Reporting Regulation (Commission Implementing Regulation (EU) 2018/2066 of 19 December 2018 on the monitoring and reporting of greenhouse gas emissions pursuant to Directive 2003/87/EC of the European Parliament and of the Council and amending Commission Regulation (EU) No 601/2012